Math 261-001
Quiz 8 Solutions, October 30

1. **Final answer:** $\boxed{42\pi}$.

   Parametrize the graph by $\vec{r}(x,y) = (x, y, xy)$ over the disk $D : x^2 + y^2 \le 15$. Its tangent vectors and area factor are

   $$\vec{r}_x = (1, 0, y), \qquad \vec{r}_y = (0, 1, x),$$
   $$\vec{n} = \vec{r}_x \times \vec{r}_y = (-y, -x, 1), \quad \|\vec{n}\| = \sqrt{1 + x^2 + y^2}.$$

   Convert the double integral over D to polar coordinates, with bounds

   $$\begin{cases} 0 \le \theta \le 2\pi, \\ 0 \le r \le \sqrt{15}. \end{cases}$$

   Thus

   $$\begin{aligned} \text{Area}(S) &= \iint_D \sqrt{1 + x^2 + y^2}\, dA \\ &= \int_0^{2\pi} \int_0^{\sqrt{15}} r\sqrt{1 + r^2}\, dr\, d\theta \\ &= \frac{2\pi}{3}\left[(1 + r^2)^{3/2}\right]_0^{\sqrt{15}} = \frac{2\pi}{3}(64 - 1) = 42\pi. \end{aligned}$$

   Here the factor r comes from $dA = r\, dr\, d\theta$. A direct polar parametrization of the surface gives the same area factor:

   $$\vec{f}(r, \theta) = (r\cos\theta, r\sin\theta, r^2\cos\theta\sin\theta), \qquad \|\vec{f}_r \times \vec{f}_\theta\| = r\sqrt{1 + r^2}.$$

2. **Final answer:** $\boxed{35\pi}$.

   The parametrization of the curve in question is $\vec{r}(t) = (t, 3t/4)$ with $2 \le t \le 4$. A parametrization of the curved lampshade is

   $$\vec{f}(t, \theta) = \left(t\cos\theta, \frac{3t}{4}, -t\sin\theta\right), \qquad \begin{cases} 2 \le t \le 4, \\ 0 \le \theta \le 2\pi. \end{cases}$$

   Its tangent vectors give

   $$\vec{f}_t = \left(\cos\theta, \frac{3}{4}, -\sin\theta\right), \qquad \vec{f}_\theta = (-t\sin\theta, 0, -t\cos\theta),$$
   $$\vec{n} = \vec{f}_\theta \times \vec{f}_t = \left(\frac{3t}{4}\cos\theta, -t, -\frac{3t}{4}\sin\theta\right), \quad \|\vec{n}\| = \frac{5t}{4}.$$

   This order gives an outward normal. The surface density becomes $\delta(\vec{f}(t, \theta)) = 3t/4$, so

   $$\begin{aligned} M &= \iint_S \sigma\, dS = \int_0^{2\pi} \int_2^4 \frac{3t}{4}\frac{5t}{4}\, dt\, d\theta \\ &= \frac{15\pi}{8}\left[\frac{t^3}{3}\right]_2^4 = \frac{5\pi}{8}(64 - 8) = 35\pi. \end{aligned}$$

3. **Final answers:**

$$\boxed{\text{(a) } \vec{r}_\phi \times \vec{r}_\theta}, \qquad \boxed{\text{(b) } \vec{f}_\theta \times \vec{f}_t}, \qquad \boxed{\text{(c) } \vec{r}_u \times \vec{r}_v}.$$

In each diagram, the two tangent vectors and the normal $\vec{n}$ start at the marked point P. Arrow lengths are rescaled for readability.

(a) For the ellipsoid,

$$\vec{r}_\phi \times \vec{r}_\theta = \left(3\sin^2\phi\cos\theta, 2\sin^2\phi\sin\theta, 6\sin\phi\cos\phi\right).$$

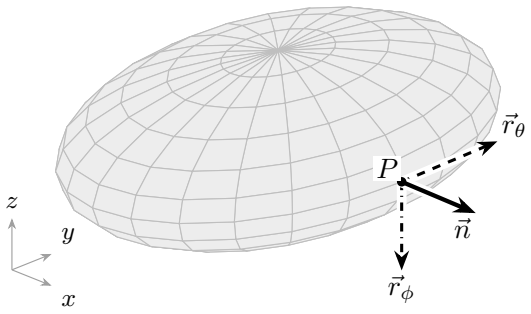
Let $f(x,y,z) = x^2/4 + y^2/9 + z^2$. Then the normal vector can be seen to be

$$\vec{r}_\phi \times \vec{r}_\theta = 3\sin\phi\,\nabla f(\vec{r}(\theta,\phi)).$$

Since $0 < \phi < \pi$, the sine is positive, and ∇f points outward from the ellipsoid, which is the surface $f = 1$. At $(\theta,\phi) = (0,\pi/2)$, the marked point is $P = (2,0,0)$, and

$$\vec{r}_\phi = (0,0,-1),$$
$$\vec{r}_\theta = (0,3,0),$$
$$\vec{n} = (3,0,0).$$

The normal points along the positive x-axis, outward from the ellipsoid.



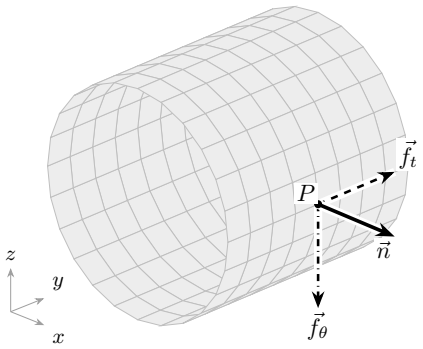
(b) For the cylinder around the y-axis,

$$\vec{f}_\theta = (-\sin\theta, 0, -\cos\theta), \qquad \vec{f}_t = (0,1,0), \qquad \vec{f}_\theta \times \vec{f}_t = (\cos\theta, 0, -\sin\theta).$$

This is exactly the radial vector from the y-axis to the surface point. At $(t,\theta) = (1,0)$, the marked point is $P = (1,1,0)$, and

$$\vec{f}_\theta = (0,0,-1),$$
$$\vec{f}_t = (0,1,0),$$
$$\vec{n} = (1,0,0).$$

The normal points away from the axis; both tangent vectors lie in the tangent plane to the curved wall.



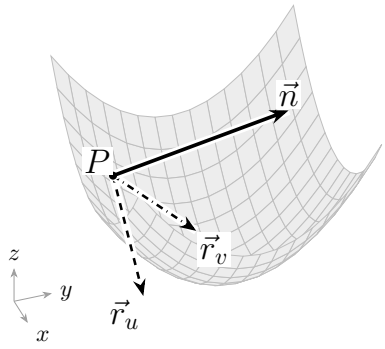
(c) For the graph $\vec{r}(u,v) = (u,v,u^2+v^2)$,

$$\vec{r}_u = (1,0,2u), \qquad \vec{r}_v = (0,1,2v), \qquad \vec{r}_u \times \vec{r}_v = (-2u,-2v,1).$$

Its positive third component makes the normal upward. At $(u,v) = (-1/2,-1/2)$, the marked point is $P = (-1/2,-1/2,1/2)$, and

$$\vec{r}_u = (1,0,-1),$$
$$\vec{r}_v = (0,1,-1),$$
$$\vec{n} = (1,1,1).$$

The diagram shows a bounded patch of the graph. The upward normal is perpendicular to the surface, not necessarily vertical.



4. **Final answers:**

$$\boxed{\text{(a) } \vec{b}}, \qquad \boxed{\text{(b) } s+t \le 1}.$$

Since $\vec{PQ} = Q - P = (2,0,0)$ and $\vec{PR} = R - P = (0,3,3)$, the affine parametrization is

$$\vec{a}(s,t) = s\vec{PQ} + t\vec{PR} + P = (2s, 3t, 1+3t).$$

With the whole unit square as its domain, its four corners map to P, Q, R, and $Q+R-P = (2,3,4)$. Therefore its image is a parallelogram, not just the given triangle.

For the ruled parametrization,

$$\vec{b}(s,t) = t\big((s\vec{PQ} + P) - R\big) + R = (2st, 3-3t, 4-3t),$$

the point $s\vec{PQ} + P = (1-s)P + sQ$ runs along the edge PQ. As t varies from 0 to 1, the ruling joins R to that point. These rulings cover exactly the given triangle. The collapsed edge $t = 0$ maps to the vertex R.

To correct the domain of $\vec{a}$, we add $s + t \le 1$, giving the bounds

$$\begin{cases} 0 \le s \le 1, \\ 0 \le t \le 1 - s. \end{cases}$$

The shaded parameter triangle maps to the given triangle; the remaining half of the square is the extra part of the parallelogram.

